WINNER RANKINGS

1	Horseshoe False Discovery Rate Control via Frequentist Thresholding Im ◀	65%
2	T-JEPA Tabular Masking with TF-Gene Interaction Structure Prevents Co-	63%
3	Inverted JEPA Masking Aligns Pretext Directionality with TF-to-Gene Ca	62%
4	GMM-Anchored JEPA Pretraining Grounds W_init in Regulatory Logic Beyon	62%
5	Conditional Independence Masking in JEPA Targets Suppresses Co-express	60%

Horseshoe False Discovery Rate Control via Frequentist Thresholding Improves DREAM5 AUPR on Small-n Networks

Status: Winner · **Domain:** Bayesian Sparse Regression / Gene Regulatory Network Inference · **Generation:** 0 · **Score:** 65%

Abstract

Standard horseshoe regression applied in Stage 2 of GRN inference produces posterior means as regulatory weight estimates, but posterior means are not sparse—they require an ad hoc thresholding rule to generate the ranked edge list needed for AUPR computation. For small-n networks like net2 (S.aureus, n=160), miscalibrated thresholding inflates false positives and degrades AUPR despite the horseshoe's theoretical sparsity properties. Recent literature establishes that integrating frequentist false discovery rate (FDR) control with horseshoe posteriors through a knockoff-inspired thresholding procedure yields both exact sparsity and calibrated edge selection [35] , while horseshoe+ extensions achieve phase transitions in signal recovery superior to standard horseshoe [24] . We hypothesize that replacing posterior-mean-based ranking with horseshoe-FDR-controlled thresholding in Stage 2 will improve AUPR on net2 by reducing false positives while maintaining sensitivity to true regulatory edges. The approach applies a frequentist-assisted horseshoe (FLASH) decision rule to select edges at a target FDR ≤ 0.20 before constructing the ranked prediction list, transforming the continuous weight distribution into a calibrated sparse prediction. Expected outcomes include net2 AUPR increase from 0.0164 to ≥0.020 (+22%) and maintained or improved AUPR on net3.

1. INTRODUCTION & SIGNIFICANCE

Paper [35] introduces the frequentist-assisted horseshoe (FLASH) approach that combines horseshoe posteriors with FDR control thresholding, solving the long-standing problem that horseshoe posterior means lack exact sparsity—a critical limitation for GRN inference where edge ranking directly determines AUPR. Paper [24] demonstrates that horseshoe+ and related heavy-tailed variants achieve superior variable selection with identifiability thresholds explicitly superior to standard horseshoe in high-dimensional sparse settings. The gap is that no existing two-stage GRN inference system applies FDR-controlled thresholding to convert horseshoe posterior distributions into calibrated sparse predictions for AUPR benchmarking.

2. HYPOTHESIS STATEMENT

> *Applying frequentist-assisted horseshoe FDR control (target FDR ≤ 0.20) to Stage 2 weight estimates on net2 will improve AUPR by at least 15% relative (from 0.0164 to ≥0.0189) compared to standard posterior-mean ranking, by reducing false positive edges in the top-100 predictions.*

3. BACKGROUND & LITERATURE REVIEW

- **False Discovery Rate Control via Frequentist-assisted Horseshoe** — Liang et al. (2025)
- **Robust Bayesian high-dimensional variable selection and inference with the horseshoe family of priors** — Fan et al. (2025)
- **Horseshoe Priors and MDP** — Polson et al. (2026)
- **Heavy-tailed and Horseshoe priors for regression and sparse Besov rates** — Agapiou et al. (2025)
- **Censored Graphical Horseshoe: Bayesian sparse precision matrix estimation with censored and missing data** — Mai et al. (2026)
- **Efficient inference of dynamic gene regulatory networks using discrete penalty** — Ravikumar et al. (2025)
- **Machine Learning Methods for Gene Regulatory Network Inference** — Hegde et al. (2025)
- **Horseshoe priors for edge-preserving linear Bayesian inversion** — Uribe et al. (2022)
- *+ 19 additional papers — see Section 8.*

4. METHODOLOGY

After Stage 2 horseshoe regression converges, apply the FLASH procedure from paper [35] to each TF-gene weight posterior: compute a p-value analog from the horseshoe posterior, apply Benjamini-Hochberg correction at target FDR ≤ 0.20, and use the resulting selected set to construct a ranked prediction list (selected edges ranked by |posterior mean|, non-selected edges ranked below). Compare AUPR to standard posterior-mean ranking on all four DREAM5 networks.

Experimental Phases

1. Implement the FLASH thresholding rule: for each (TF, gene) pair, compute a horseshoe-based test statistic from the ratio of posterior mean to posterior standard deviation of the weight w_{ij}; convert to a two-sided p-value using a half-Cauchy calibration as described in paper [35] .
2. Apply Benjamini-Hochberg FDR correction at $q \in \{0.10, 0.15, 0.20, 0.30\}$ to obtain a sparse selected edge set; tune q on a held-out 20% validation split of each network's gold standard edges.
3. Construct the final ranked prediction list: selected edges ranked by |posterior mean|, followed by non-selected edges ranked by posterior probability of inclusion; compute AUPR on the full DREAM5 gold standard.
4. Compare AUPR of FLASH-thresholded predictions to: (a) standard posterior-mean ranking, (b) posterior median ranking, (c) horseshoe+ posterior mean ranking (implementing a heavier-tailed local prior $\lambda_j \sim$ Student-t(1) instead of half-Cauchy) as per paper [24] .
5. Specifically on net2 (n=160), analyze the precision-recall curve shape: FLASH should improve precision in the high-confidence region (top-100 edges) while maintaining recall; measure AUPR over the full curve and the partial AUPR at recall ≤ 0.20.
6. Report calibration: compare declared FDR at threshold q to empirical false discovery proportion in the top-k predictions using the gold standard; confirm FLASH is better calibrated than posterior-mean ranking.

5. EXPECTED OUTCOMES

Predicted Impact: Net2 AUPR improvement from 0.0164 to ≥0.020 (+22%); improved top-100 prediction precision from estimated 30% to ≥40% on net2; net3 AUPR improvement from ~0.090 to ≥0.095 (partial improvement, with Stage 1 fix needed for full GENIE3 beat); AUROC for expression-only baseline ≥ 0.70 maintained.

Success Criteria: Net2 AUPR ≥ 0.0189 (15% relative improvement over 0.0164); empirical FDR in top-100 predictions ≤ 0.25 on net2; FLASH calibration error (|declared FDR - empirical FDP|) < 0.05; no degradation of net1 or net4 AUPR by more than 5% relative; Stage 2 inference time increases by < 10% due to FLASH post-processing.

Null Result Interpretation: If this hypothesis is false: Instead of hard BH thresholding, use the horseshoe local shrinkage factor $(1 - \kappa_{ij})$ as a continuous multiplicative correction to posterior means (the composite score from hypothesis a2a37746), avoiding the calibration issues of frequentist-Bayesian mixing while still down-weighting shrunk estimates.

6. RISK ANALYSIS & MITIGATION

Fatal Assumptions

- ☠ FLASH-derived p-value analogs from horseshoe posteriors are well-calibrated under the null for TF-gene pairs with no regulatory relationship — if false: the BH correction selects a severely biased subset, and the ranked list is worse than posterior-mean ranking alone.
- ☠ The selected edge set at FDR ≤ 0.20 contains enough edges to compute a meaningful AUPR with demonstrable improvement over baseline — if the selected set is very small (e.g., <50 edges per TF), the AUPR computation is dominated by interpolation artifacts.

Weakest Assumption: That the FLASH frequentist-horseshoe procedure produces calibrated FDR estimates when applied to adaptive horseshoe posteriors in the GRN inference setting — the mismatch between Bayesian shrinkage and frequentist null hypothesis testing fundamentally undermines the validity of the FDR control claim.

Key Risks

- FLASH FDR control requires a valid null distribution; horseshoe posteriors do not have a direct frequentist null, so converting them to p-value analogs via tail probability involves distributional assumptions that are misspecified when the prior is itself adaptive (horseshoe), inflating or deflating effective FDR estimates.
- With a target FDR of 0.20 on DREAM5 networks where true edge prevalence is ~1–5%, BH correction at FDR ≤ 0.20 will select very few edges (or none below threshold), causing the 'selected set' to be too small for meaningful AUPR calculation, especially for top-100 predictions.
- The 15% relative AUPR gain claim (0.0164 → 0.0189) is small in absolute terms and well within the noise level of DREAM5 evaluation; a minor change in how tied scores are broken or which edges appear at rank 100 could explain the entire improvement.
- Hard thresholding non-selected edges below selected edges creates a discrete jump in the ranking that can hurt AUPR by artificially compressing the informative part of the precision-recall curve.

Minimal Validation Experiment

On net2, compute the horseshoe posterior for all TF-gene pairs and plot the empirical distribution of p-value analogs for known non-edges vs. known edges (from gold standard). Takes ~2 days. If the p-value distributions strongly overlap (AUC of p-value analog < 0.60 on known edge labels), FLASH FDR control cannot produce calibrated selection and the hypothesis fails.

Competing Mechanisms

- ⇄ Alternative 1 (Ranking Invariance): AUPR is determined by the relative ranking of all (TF, gene) pairs, not by the absolute threshold — if true, any monotone transformation of posterior means would produce identical AUPR, and FLASH thresholding (which reorders the boundary region) would show no statistically significant AUPR improvement over posterior-mean ranking.

- ⇄ Alternative 2 (Global Shrinkage Miscalibration): The dominant source of false positives on net2 is not the thresholding rule but the global shrinkage parameter τ being too large (under-shrinking), meaning improving τ selection (e.g., via empirical Bayes τ estimation) would yield larger AUPR gains than FLASH thresholding while leaving the relative ranking largely unchanged.

Discriminating Experiment

On net2, run Stage 2 to convergence, then generate three prediction lists: (A) FLASH FDR-thresholded at q=0.20, (B) standard posterior-mean ranking, (C) empirical Bayes τ-optimized ranking (tuning τ on validation split). Compute full AUPR and partial AUPR at recall ≤ 0.20 (which captures precision in the top predictions). If THIS hypothesis is correct: AUPR(A) > AUPR(B) and the precision-recall curve of A dominates B at low recall. If Alternative 1 is correct: AUPR(A) ≈ AUPR(B) within noise. If Alternative 2 is correct: AUPR(C) > AUPR(A) > AUPR(B), with C showing the largest gain.

7. LIMITATIONS & ASSUMPTIONS

Key Assumptions (most → least confident)

1. Assumption [85%]: Horseshoe posterior standard deviations are well-estimated by the variational Bayes or EM approximation used in Stage 2, enabling reliable FLASH test statistics — Risk: VI approximations systematically underestimate posterior variance, making the FLASH p-values anti-conservative.
2. Assumption [75%]: The DREAM5 gold standard edge labels provide sufficient positives in net2 to tune FDR threshold q on a held-out 20% validation split without overfitting — Risk: net2 has few known edges, making the validation split too small for stable q tuning.
3. Assumption [60%]: FDR control at q=0.20 aligns with the precision-recall operating point that maximizes net2 AUPR — Risk: the AUPR-maximizing operating point may be at a very different FDR level, requiring extensive q tuning that itself overfits the gold standard.
4. Assumption [40%]: Improving precision in the top-100 predictions (via FLASH) will improve the full-curve AUPR metric used by DREAM5 scoring — Risk: DREAM5 AUPR integrates over the full recall range, and FLASH may improve high-precision low-recall performance at the cost of mid-recall performance, leaving full-curve AUPR unchanged.

Major Assumptions — require revision if false

- ⚠ The horseshoe posterior variance faithfully encodes uncertainty about edge existence rather than reflecting the prior's adaptive shrinkage behavior — if false: the FDR analog is actually measuring prior-induced shrinkage, not data evidence.
- ⚠ The 15% relative gain target is achievable with FDR-based re-ranking rather than requiring a better Stage 1 initialization — if false: Stage 2 improvements are ceiling-limited by W_init quality, and FDR filtering cannot compensate for poor initial weights.

8. EVIDENCE SUMMARY

DIMENSION	SCORE	WEIGHT
EVIDENCE GROUNDING	70%	35%
Feasibility	72%	30%
Impact Potential	55%	25%
Novelty	48%	10%
Composite	**65%**	weighted

Why this won: This hypothesis is distinct because it targets the Stage 2 output conversion problem—posterior means to ranked predictions—rather than Stage 1 initialization. The FLASH approach from paper [35] directly addresses the exact sparsity gap in horseshoe regression, which is critical for AUPR computation in sparse biological networks. Net2 with n=160 is where Bayesian calibration matters most (as noted in the constraints), making this the highest-leverage network for this intervention. The approach is computationally trivial (post-processing on MPS after convergence) and requires no changes to training, making it the most feasible hypothesis in the set.

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